



GE Medical Systems

Technical Publication

Direction 2224901-100

Revision 05

**GE Medical Systems
SmartScore Option Manual for CT/i &
LightSpeed**

Copyright © 1999–2003 by General Electric Company, Inc.
All rights reserved.



LEGAL NOTES

TRADEMARKS

Adobe, the Adobe logo, Acrobat, the Acrobat logo, Exchange, and PostScript are trademarks of Adobe Systems Incorporated or its subsidiaries and may be registered in certain jurisdictions.

Microsoft is a registered trademark and Windows is a trademark of Microsoft Corporation.

All other products and their name brands are trademarks of their respective holders.

COPYRIGHTS

All Material Copyright © 1999–2003 by General Electric Company, Inc. All rights reserved.

IMPORTANT PRECAUTIONS

LANGUAGE

WARNING

- THIS SERVICE MANUAL IS AVAILABLE IN ENGLISH ONLY.
- IF A CUSTOMER'S SERVICE PROVIDER REQUIRES A LANGUAGE OTHER THAN ENGLISH, IT IS THE CUSTOMER'S RESPONSIBILITY TO PROVIDE TRANSLATION SERVICES.
- DO NOT ATTEMPT TO SERVICE THE EQUIPMENT UNLESS THIS SERVICE MANUAL HAS BEEN CONSULTED AND IS UNDERSTOOD.
- FAILURE TO HEED THIS WARNING MAY RESULT IN INJURY TO THE SERVICE PROVIDER, OPERATOR OR PATIENT FROM ELECTRIC SHOCK, MECHANICAL OR OTHER HAZARDS.

AVERTISSEMENT

- CE MANUEL DE MAINTENANCE N'EST DISPONIBLE QU'EN ANGLAIS.
- SI LE TECHNICIEN DU CLIENT A BESOIN DE CE MANUEL DANS UNE AUTRE LANGUE QUE L'ANGLAIS, C'EST AU CLIENT QU'IL INCOMBE DE LE FAIRE TRADUIRE.
- NE PAS TENTER D'INTERVENTION SUR LES ÉQUIPEMENTS TANT QUE LE MANUEL SERVICE N'A PAS ÉTÉ CONSULTÉ ET COMPRIS.
- LE NON-RESPECT DE CET AVERTISSEMENT PEUT ENTRAÎNER CHEZ LE TECHNICIEN, L'OPÉRATEUR OU LE PATIENT DES BLESSURES DUES À DES DANGERS ÉLECTRIQUES, MÉCANIQUES OU AUTRES.

WARNUNG

- DIESES KUNDENDIENST-HANDBUCH EXISTIERT NUR IN ENGLISCHER SPRACHE.
- FALLS EIN FREMDER KUNDENDIENST EINE ANDERE SPRACHE BENÖTIGT, IST ES AUFGABE DES KUNDEN FÜR EINE ENTSPRECHENDE ÜBERSETZUNG ZU SORGEN.
- VERSUCHEN SIE NICHT, DAS GERÄT ZU REPARIEREN, BEVOR DIESES KUNDENDIENST-HANDBUCH NICHT ZU RATE GEZOGEN UND VERSTANDEN WURDE.
- WIRD DIESE WARNUNG NICHT BEACHTET, SO KANN ES ZU VERLETZUNGEN DES KUNDENDIENSTTECHNIKERS, DES BEDIENERS ODER DES PATIENTEN DURCH ELEKTRISCHE SCHLÄGE, MECHANISCHE ODER SONSTIGE GEFAHREN KOMMEN.

AVISO

- ESTE MANUAL DE SERVICIO SÓLO EXISTE EN INGLÉS.
- SI ALGÚN PROVEEDOR DE SERVICIOS AJENO A GEMS SOLICITA UN IDIOMA QUE NO SEA EL INGLÉS, ES RESPONSABILIDAD DEL CLIENTE OFRECER UN SERVICIO DE TRADUCCIÓN.
- NO SE DEBERÁ DAR SERVICIO TÉCNICO AL EQUIPO, SIN HABER CONSULTADO Y COMPRENDIDO ESTE MANUAL DE SERVICIO.
- LA NO OBSERVANCIA DEL PRESENTE AVISO PUEDE DAR LUGAR A QUE EL PROVEEDOR DE SERVICIOS, EL OPERADOR O EL PACIENTE SUFRAN LESIONES PROVOCADAS POR CAUSAS ELÉCTRICAS, MECÁNICAS O DE OTRA NATURALEZA.

ATENÇÃO

- ESTE MANUAL DE ASSISTÊNCIA TÉCNICA SÓ SE ENCONTRA DISPONÍVEL EM INGLÊS.
- SE QUALQUER OUTRO SERVIÇO DE ASSISTÊNCIA TÉCNICA, QUE NÃO A GEMS, SOLICITAR ESTES MANUAIS NOUTRO IDIOMA, É DA RESPONSABILIDADE DO CLIENTE FORNECER OS SERVIÇOS DE TRADUÇÃO.
- NÃO TENHA TENTADO REPARAR O EQUIPAMENTO SEM TER CONSULTADO E COMPREENDIDO ESTE MANUAL DE ASSISTÊNCIA TÉCNICA.
- O NÃO CUMPRIMENTO DESTA AVISO PODE POR EM PERIGO A SEGURANÇA DO TÉCNICO, OPERADOR OU PACIENTE DEVIDO A CHOQUES ELÉTRICOS, MECÂNICOS OU OUTROS.

AVVERTENZA

- IL PRESENTE MANUALE DI MANUTENZIONE È DISPONIBILE SOLTANTO IN INGLESE.
- SE UN ADDETTO ALLA MANUTENZIONE ESTERNO ALLA GEMS RICHIEDE IL MANUALE IN UNA LINGUA DIVERSA, IL CLIENTE È TENUTO A PROVVEDERE DIRETTAMENTE ALLA TRADUZIONE.
- SI PROCEDA ALLA MANUTENZIONE DELL'APPARECCHIATURA SOLO DOPO AVER CONSULTATO IL PRESENTE MANUALE ED AVERNE COMPRESO IL CONTENUTO.
- NON TENERE CONTO DELLA PRESENTE AVVERTENZA POTREBBE FAR COMPIERE OPERAZIONI DA CUI DERIVINO LESIONI ALL'ADDETTO ALLA MANUTENZIONE, ALL'UTILIZZATORE ED AL PAZIENTE PER FOLGORAZIONE ELETTRICA, PER URTI MECCANICI OD ALTRI RISCHI.

警告

このサービスマニュアルには英語版しかありません。

GEMS以外でサービスを担当される業者が英語以外の言語を要求される場合、翻訳作業はその業者の責任で行うものとさせていただきます。

このサービスマニュアルを熟読し理解せずに、装置のサービスを行わないで下さい。

この警告に従わない場合、サービスを担当される方、操作員あるいは患者さんが、感電や機械的又はその他の危険により負傷する可能性があります。

注意:

本维修手册仅存有英文本。
非 GEMS 公司的维修员要求非英文本的维修手册时，
客户需自行负责翻译。
未详细阅读和完全了解本手册之前，不得进行维修。
忽略本注意事项会对维修员，操作员或病人造成触
电，机械伤害或其他伤害。

DAMAGE IN TRANSPORTATION

All packages should be closely examined at time of delivery. If damage is apparent write "Damage In Shipment" on ALL copies of the freight or express bill BEFORE delivery is accepted or "signed for" by a GE representative or hospital receiving agent. Whether noted or concealed, damage MUST be reported to the carrier immediately upon discovery, or in any event, within 14 days after receipt, and the contents and containers held for inspection by the carrier. A transportation company will not pay a claim for damage if an inspection is not requested within this 14 day period.

Call Traffic and Transportation, Milwaukee, WI (414) 785 5052 or 8*323 5052 immediately after damage is found. At this time be ready to supply name of carrier, delivery date, consignee name, freight or express bill number, item damaged and extent of damage.

Complete instructions regarding claim procedure are found in Section S of the Policy And Procedures Bulletins.

14 July 1993

CERTIFIED ELECTRICAL CONTRACTOR STATEMENT

All electrical Installations that are preliminary to positioning of the equipment at the site prepared for the equipment shall be performed by licensed electrical contractors. In addition, electrical feeds into the Power Distribution Unit shall be performed by licensed electrical contractors. Other connections between pieces of electrical equipment, calibrations and testing shall be performed by qualified GE Medical personnel. The products involved (and the accompanying electrical installations) are highly sophisticated, and special engineering competence is required. In performing all electrical work on these products, GE will use its own specially trained field engineers. All of GE's electrical work on these products will comply with the requirements of the applicable electrical codes.

The purchaser of GE equipment shall only utilize qualified personnel (i.e., GE's field engineers, personnel of third-party service companies with equivalent training, or licensed electricians) to perform electrical servicing on the equipment.

IMPORTANT...X-RAY PROTECTION

X-ray equipment if not properly used may cause injury. Accordingly, the instructions herein contained should be thoroughly read and understood by everyone who will use the equipment before you attempt to place this equipment in operation. The General Electric Company, Medical Systems Group, will be glad to assist and cooperate in placing this equipment in use.

Although this apparatus incorporates a high degree of protection against x-radiation other than the useful beam, no practical design of equipment can provide complete protection. Nor can any practical design compel the operator to take adequate precautions to prevent the possibility of any persons carelessly exposing themselves or others to radiation.

It is important that anyone having anything to do with x-radiation be properly trained and fully acquainted with the recommendations of the National Council on Radiation Protection and Measurements as published in NCRP Reports available from NCRP Publications, 7910 Woodmont

Avenue, Room 1016, Bethesda, Maryland 20814, and of the International Commission on Radiation Protection, and take adequate steps to protect against injury.

The equipment is sold with the understanding that the General Electric Company, Medical Systems Group, its agents, and representatives have no responsibility for injury or damage which may result from improper use of the equipment.

Various protective materials and devices are available. It is urged that such materials or devices be used.

LITHIUM BATTERY CAUTIONARY STATEMENTS



CAUTION
Risk of
Explosion

Danger of explosion if battery is incorrectly replaced. Replace only with the same or equivalent type recommended by the manufacturer. Discard used batteries according to the manufacturer's instructions.



ATTENTION
Danger
d'Explosion

Il y a danger d'explosion s'il y a remplacement incorrect de la batterie. Remplacer uniquement avec une batterie du même type ou d'un type recommandé par le constructeur. Mettre au rebut les batteries usagées conformément aux instructions du fabricant.

Revision History

Revision	Date	Reason for change
0	Nov. 1998	Initial product release
1	March 1999	Changes made: 1.) Final Product Changes Include 2.) Chapters rearranged
2	March 2000	Changed - Updated CD-ROM Part Number in SmartScore Option Parts List Table -Chapter 2. "From a dash 2 to a dash 4." - Change the word "in" to "is", in step 1 of Chapter 7.
3	August 2001	Added "for CTi & LightSpeed." Rearranged subsections in Chapter 3. Added SmartScore 3.3 Option Installation information to Chapter 5.
4	June 2002	Updates to Chapter 5.
5	Feb. 2003	Updates to Chapter 5: CSA NPI engineering, SmartScore 3.5 & AW Linux affectivity.

Table of Contents

Preface

Publication Conventions	15
--------------------------------------	-----------

Section 1.0

Safety & Hazard Information	15
--	-----------

1.1 Text and Character Representation.....	15
1.2 Graphical Representation	16

Section 2.0

Publication Conventions	17
--------------------------------------	-----------

2.1 General Paragraph and Character Styles.....	17
2.2 Page Layout.....	17
2.3 Computer Screen Output/Input Character Styles	18

Chapter 1

Introduction.....	19
--------------------------	-----------

Section 1.0

Cardiac Calcification Theory.....	19
--	-----------

Section 2.0

Assumptions.....	19
-------------------------	-----------

Section 3.0

Manual Description	20
---------------------------------	-----------

Chapter 2

Upgrade Overview for SmartScore	21
--	-----------

Section 1.0

SmartScore Upgrade Parts List	21
--	-----------

Section 2.0

Cardiac Upgrade Checklist.....	22
---------------------------------------	-----------

Chapter 3

Hardware (Gantry) Upgrade 23

Section 1.0

Remove Covers from Gantry 23

Section 2.0

Install Opto Driver II Board and Cardiac Harness 23

Section 3.0

Install Opto-Isolator Board 27

Section 4.0

Optional Disk Drive..... 27

Chapter 4

System Functional Checks..... 29

Section 1.0

Create Test EKG Waveform 29

Section 2.0

View the Test EKG Waveform..... 30

Chapter 5

SmartScore Option (AW Software) Installation 31

Section 1.0

Introduction: “New” vs. “Old” SmartScore..... 31

Section 2.0

Summary of SmartScore Software Installation..... 31

Section 3.0

SmartScore Software Installation Preparation 32

Section 4.0

SmartScore AW Software Installation Outline 33

Section 5.0

Installation & Uninstallation Notes 33

Section 6.0	
Test and Maintenance of SmartScore	34
6.1 SmartScore Application Configuration Save.....	34
6.2 SmartScore Configuration SAVE/RESTORE Test	35
 Section 7.0	
SmartScore 3.5 and Beyond, File Structure Information.....	36
 Section 8.0	
SmartScore Maintenance Information	
(Declaration of Image Failure Error or File System Full Issues).....	36
8.1 Information	36
8.2 Procedure	37
8.2.1 For SmartScore 3.3 only (until it is replaced by 3.5)	37
8.2.2 Service Procedure – from a command unix window.....	37
8.2.2.1 Finding the capacity of the ~sdc/sscore/studies directory	37
8.2.2.2 Removal of ~sdc/sscore/studies save state files	37
 Section 9.0	
SmartScore Operation and Demo Exam Information	38
 Section 10.0	
SmartScore Error Log Information	38
 Section 11.0	
SmartScore Printing and Filming Information.....	38
 Section 12.0	
SmartScore “Special” Considerations.....	39
12.1 Read This First	39
12.2 Procedure	39
12.3 Check Scoring Configuration	40
 Chapter 6	
SmartScore (System) Interconnect.....	41
 Chapter 7	
Troubleshooting	43

Preface

Publication Conventions

Purpose: This section means to inform the reader on publication conventions used. So that the reader can identify safety and general material that is considered important by its format. This includes the interpretation of computer screen text as either input or output. There are a number of specific text and paragraph styles/conventions used within this section to accomplish this task. Please become familiar with the conventions used within this publication before proceeding.

Section 1.0

Safety & Hazard Information

1.1 Text and Character Representation

Within this publication, different paragraph and character styles have been used to indicate potential hazards. Paragraph prefixes, such as hazard, caution, danger and warning, are used to identify important safety information. Text (Hazard) styles are applied to the paragraph contents that are applicable to each specific safety statement. Words describe the type of potential hazard that may be encountered and are placed immediately before the paragraph it modifies. Safety information will normally include:

- Type of potential Hazard
- Nature of potential injury
- Causative condition
- How to avoid or correct the causative condition

EXAMPLES OF HAZARD STATEMENTS USED

A few examples are provided that have been adapted from GEMS' global document standard (2119696-100). They include paragraph prefixes and modified text styles.



CAUTION
Pinch Points
Loss of Data
Sharp Objects

Caution is used when a hazard exists which can or could cause minor injury to self or others if instructions are ignored. They include for example:

- **Loss of critical patient data**
- **Crush or pinch points**
- **Sharp objects**



DANGER
EXCESSIVE
VOLTAGE
CRUSH
POINT

DANGER IS USED WHEN A HAZARD EXISTS WHICH WILL CAUSE SEVERE PERSONAL INJURY OR DEATH IF INSTRUCTIONS ARE IGNORED. THEY CAN INCLUDE:

- **ELECTROCUTION**
- **CRUSHING**
- **RADIATION**



**WARNING
ROTATING
EQUIPMENT
BARE WIRES**

WARNING IS USED WHEN A HAZARD EXISTS WHICH COULD OR CAN CAUSE SERIOUS PERSONAL INJURY OR DEATH IF INSTRUCTIONS ARE IGNORED. THEY CAN INCLUDE:

- Potential for shock
- Exposed wires
- Failure to Tag and lockout system power could allow for un-command motion.



**NOTICE
Equipment
Damage
Possible**

Notice is used when a hazard is present that can cause property damage but has absolutely no personal injury risk. They can include:

- Disk drive will crash
- Internal mechanical damage, such as to the x-ray tube
- Coasting the rotor through resonance.

It's important that the reader not ignore hazard statements in this document.

1.2 Graphical Representation

Important information will always be preceded by the exclamation point  contained within a triangle, as seen throughout this chapter. In addition to text, several different graphical icons (symbols) may be used to make you aware of specific types of hazards that could cause harm.

ELECTRICAL



LASER



MECHANICAL



HEAT



RADIATION



PINCH



Some others make you aware of specific procedures that should be followed.

AVOID STATIC ELECTRICITY



TAG AND LOCK OUT



WEAR EYE PROTECTION



Section 2.0 Publication Conventions

2.1 General Paragraph and Character Styles

Prefixes are used to highlight important non-safety related information. Paragraph prefixes (such as Purpose, Example, Comment and Note) are used to identify important but non-safety related information. Text styles are also applied to text within each paragraph modified by the specific prefix.

EXAMPLES OF PREFIXES USED FOR GENERAL INFORMATION

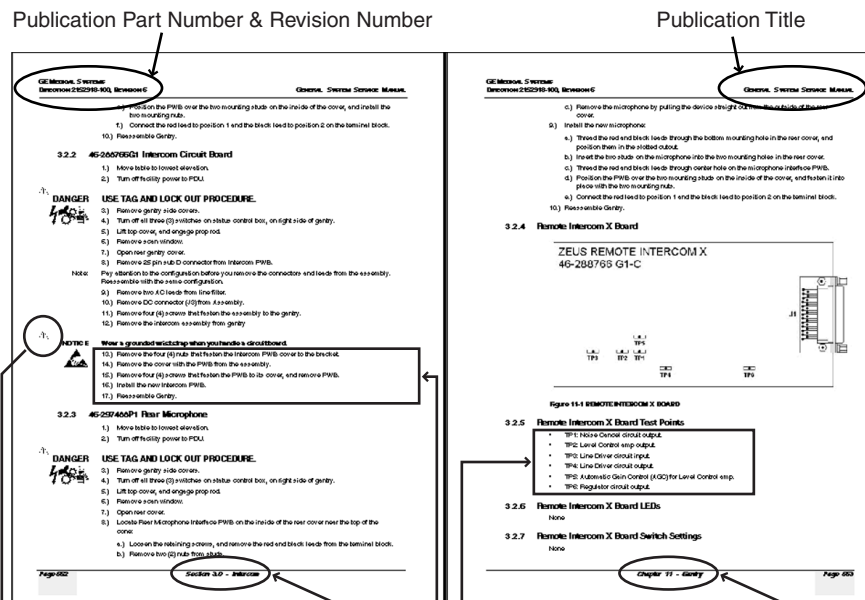
Purpose: Introduces and provides meaning as to the information contained within the chapter, section or subsection, Such as used at the beginning this chapter for example.

Note: Conveys information that should be considered important to the reader.

Example: Used to make the reader aware that the paragraph(s) that follow are examples of information possibly stated previously.

Comment: Represents “additional” information that may or may not be relevant.

2.2 Page Layout



The current section and its title are always shown in the footer of the left (even) page.

An exclamation point in a triangle is used to indicate important information to the user.

Paragraphs preceded by **Alphanumeric** (e.g. numbers) characters is information that must be followed in a **specific order**.

The current chapter and its title are always shown in the footer of the right (odd) page.

Paragraphs preceded by **symbols** is (e.g. bullets) is information that has **no specific order**.

Headers and footers in this publication are designed to allow you to quickly identify your location. The document's part number and revision number appears in every header on every page. Odd

numbered page footers indicate the current chapter, its title and current page number. Even page footers show the current section and its title, as well current page number.

2.3 Computer Screen Output/Input Character Styles

Within this publication different character styles are used to indicate computer input and output text. Character (input, output, and variable) styles are used and applied to the text within a paragraph so as to indicated direction. Computer screen output and input is also formatted using mono (fixed width) spaced fonts.

Example: This paragraph denotes computer screen fixed output. It's output is fixed
Fixed Output from the sense that it does not vary from application to application. It's
the most commonly used style used to indicate filenames, paths and text.

Example: *This paragraph denotes computer screen output that is variable. Its output
Variable Output varies from application to application. Variable output is sometimes found
placed between greater than and lesser than operators. For example:
<variable_ouput>*

Example: This paragraph denotes fixed input. It's typed input that will not vary
Fixed Input from application to application. Fixed text the user is required to supply
as input.

Example: *This paragraph denotes computer input that can vary from application to
Variable Input application. Variable text the user is required to supply as input.
Variable input sometimes is placed between greater than and lesser than
operators. For example: <variable_input>. In these cases, the (<>)
operators are dropped prior to input. Exceptions are noted in the text.*

End of Section

Chapter 1

Introduction

Follow the instructions in this document to upgrade the HSA,CT/i, or LightSpeed Gantry to accommodate an EKG machine, for use during Cardiac studies.

Section 1.0

Cardiac Calcification Theory

During the Cardiac study, the technologist attaches EKG monitors to the patient, and acquires a Helical study of the chest area. Upon completion of the study, the Advantage Windows cardiac software uses the EKG data to extract images from the study that show the heart at rest. Physicians can examine the resting heart images for signs of calcification, or other pathology.

The purpose of this interface is to provide isolation between the CT Gantry and the EKG. The interface provides for power, signal and ground isolation. The X-RAY ON signal appearing between A3J2A21 and A3J2A22 disables current flow into the opto Isolator on Pin 2 which returns through Pin 3 of the Opto Driver located on the STC backplane. The Opto Isolator output driver is a transistor within the Opto Isolator which is switched into conduction by light emitted by the internal LED, when current flows from the Opto driver. The switch is closed with no X-RAY ON signal. The switch opens when X-RAY ON is high. During normal operation, the EKG continually test for connection integrity.

Section 2.0

Assumptions

This publication assumes the facility has an Advantage Windows system with version 3.1, or higher, software.

It is also recommended that this option be used on a CT/i performix system or LightSpeed. This option will work on an HSA system but is not recommended.

Section 3.0

Manual Description

Note: Systems with the following model numbers were manufactured with a SmartScore cable-ready harness in the Gantry. If the cable is already present, you will be directed to skip steps in the procedure.

- 2183250-2
- 2183255-2
- 2119734-3
- 2119732-4

In addition to this Introduction (Chapter 1), this manual contains the following:

- [Chapter 2](#): Upgrade kit parts list and upgrade flowchart/checklist
- [Chapter 3](#): Hardware Upgrade procedures
- [Chapter 4](#): System Functional Checks
- [Chapter 5](#): Software Installation Information
- [Chapter 6](#): System Interconnects
- [Chapter 7](#): Troubleshooting

Chapter 2

Upgrade Overview for SmartScore

Section 1.0

SmartScore Upgrade Parts List

	Part Number	Description
1.)	2217017	Opto Driver II
2.)	2216888	Opto Driver to Opto-isolator Cable
3.)	2216886	Gantry to Opto-isolator Board/Cable assembly
4.)	516A770P405	Screw
5.)	46-136375P32	Hex Spacer, 3/16 HEX. 0.88 Long
6.)	46-328421P14	Screw, Pan Head 4mm, 10mm
7.)	2215053	Left Base Cover
8.)	2216584	Cardiac Coverplate
9.)	2218235	Cover - STC Cardiac option
10.)	2216892	Gantry to EKG Cable
11.)	46-302200P3	Rating Plate
12.)	2239892-100	Operator Manual
13.)	2228113-4	CD-ROM
14.)	E8007N	Starter Kit
	• E8007MG	• Box of Floppy Disks
	• E8007ME	• EKG Electrodes
15.)	E8007MH	Heartbeat Simulator

Table 2-1 SmartScore Option Parts list

	Part Number	Description
1.)	B7510S	Optional Disk Drive for Indigo
2.)	B7710JD	Optional Disk Drive for Octane

Table 2-2 Optional Disk Drive

Section 2.0

Cardiac Upgrade Checklist

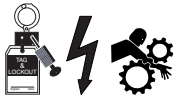
Please keep track of your progress. Place a checkmark next to each item in the following list as you complete it.

Note: Some SmartScore kits come with an optional disk drive for the OC to increase the customers image space. This optional disk drive can be installed at the same time as the gantry hardware is being installed. Also, the optional disk drive could be broken out from this installation and be installed at a later date and time depending on customer time constraints. If postponed, be aware the customers image space could be limited.



NOTICE

Remove Power, and remember to Tag and Lockout hazards.



- 1.) Remove Gantry covers
- 2.) Install Opto-Driver Board II and Cardiac harness.
- 3.) Install Gantry Base cover.
- 4.) Install Opto-Isolator board and cable.
- 5.) Unpack and install EKG machine (follow manufacturer's instructions).
- 6.) Restore System power.
- 7.) Install AW 3.1 software
- 8.) Check System Function.
- 9.) Distribute operators manual and accessories kit to customers

Chapter 3

Hardware (Gantry) Upgrade

Section 1.0

Remove Covers from Gantry

- 1.) Turn off the Mains disconnect power, A1.



NOTICE



Remove Power, and remember to Tag and Lockout hazards.

- 2.) Remove both side covers from the Gantry.
- 3.) Turn off the following Gantry Service switches:
 - 550VDC
 - 120VAC
 - Gantry Rotate (Axial Drive Enable)
- 4.) Open the Front and Rear Gantry covers.
- 5.) Remove, and set aside, the left side cover from the Gantry base.

Section 2.0

Install Opto Driver II Board and Cardiac Harness

- 1.) Refer to [Figure 3-1](#).
- 2.) Remove the STC cover to expose the STC backplane, and set aside.
- 3.) Remove the screws that fasten the Harness bracket to the bottom of the STC.
- 4.) Temporarily remove the A3-J2 and A3-J3 cables from the STC backplane.
 - a.) Remove the Opto Driver II board (also called Cardiac/Sonata board) from the upgrade kit.
 - b.) Plug the Cardiac Cable, 2216888, into J2 on the Opto Driver II board
 - c.) Orient the Opto-Driver II board with the J2 pins pointing toward the floor, and plug it into A3-J2 and A3-J3 on the backplane.
 - d.) Use hardware from the kit to fasten the Opto-isolator board to the spacers on the STC backplane.
 - e.) Plug the A3-J2 and A3-J3 cables into the corresponding connectors on the opto-isolator board.
 - f.) Use the captive hardware on A3-J2 and A3-J3 to fasten the connectors into place.

Note: Cable present at J1 only when the system has SmartView option.

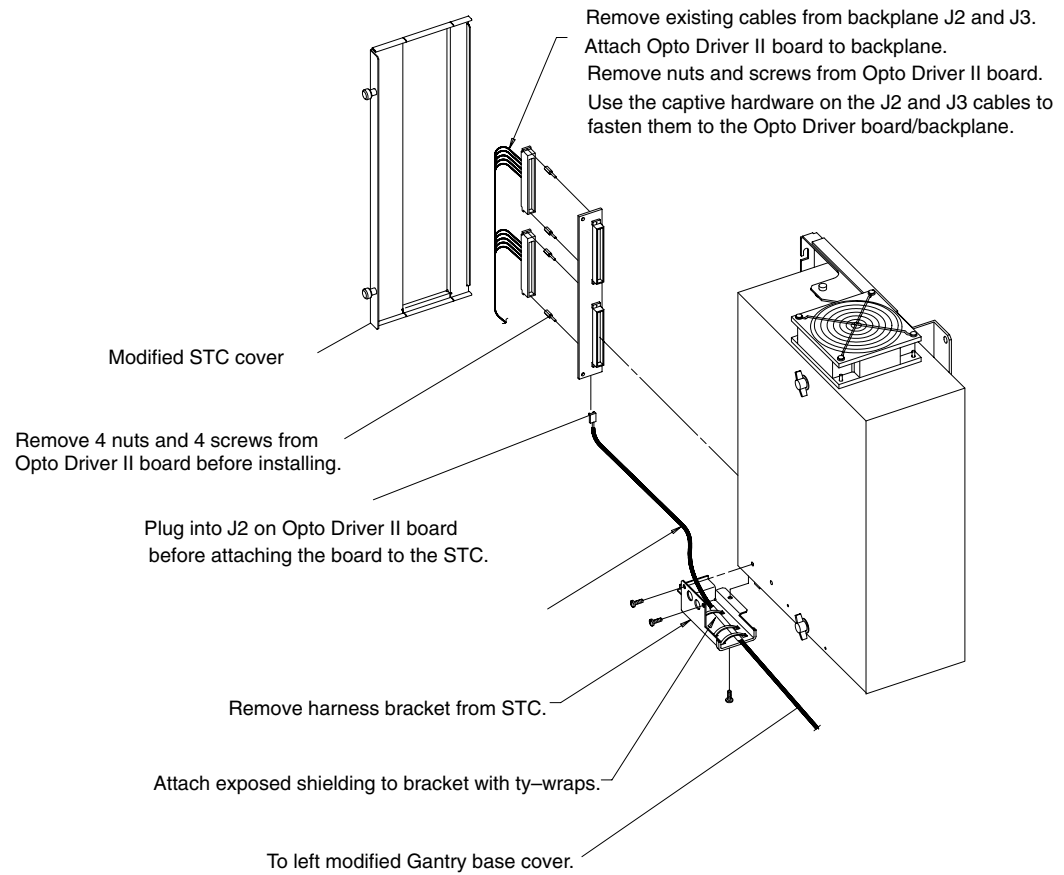


Figure 3-1 Add Opto-Driver II board to STC Backplane

- 5.) Use ty-wraps to fasten the exposed shielding on the end of the Cardiac Harness to the existing cable shields.

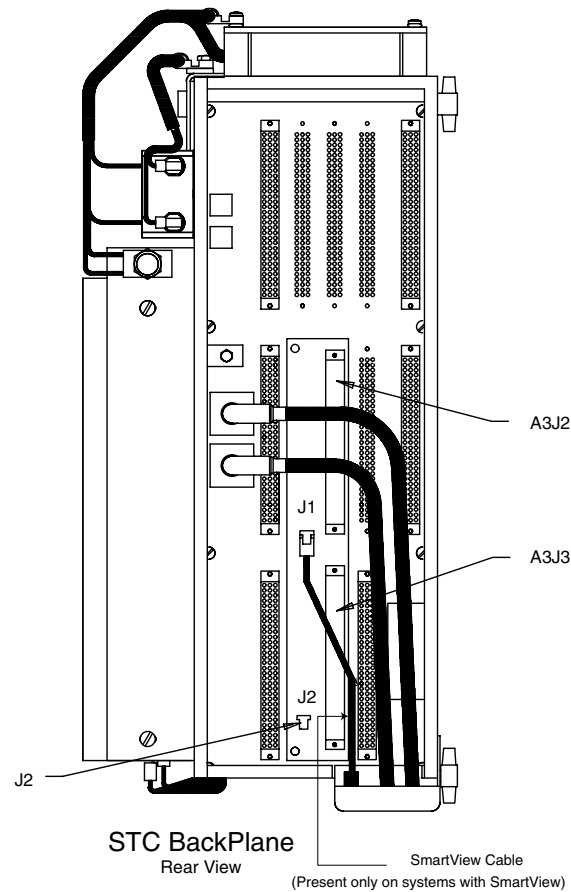


Figure 3-2 STC Backplane with the Opto Driver II Board in Place

- 6.) Reattach the Harness bracket to the bottom of the STC.
- 7.) Remove the modified STC backplane cover from the upgrade kit, and fasten it into place. (Discard the old STC cover.)

Note: Some Gantry Models already have the modified STC cover in place. The old cover is flat. The new cover has a raised center section and knurled fasteners.

- 8.) Refer to [Figure 3-3](#). Route the Cardiac harness down the rotating base and cable router.

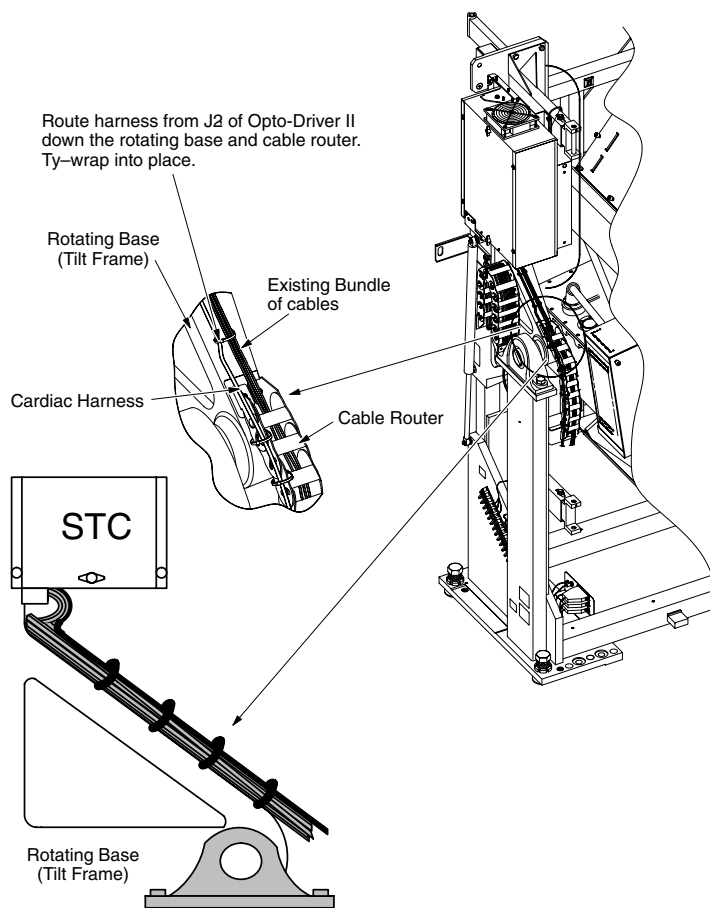


Figure 3-3 Route the Cardiac Harness down the Rotating Base

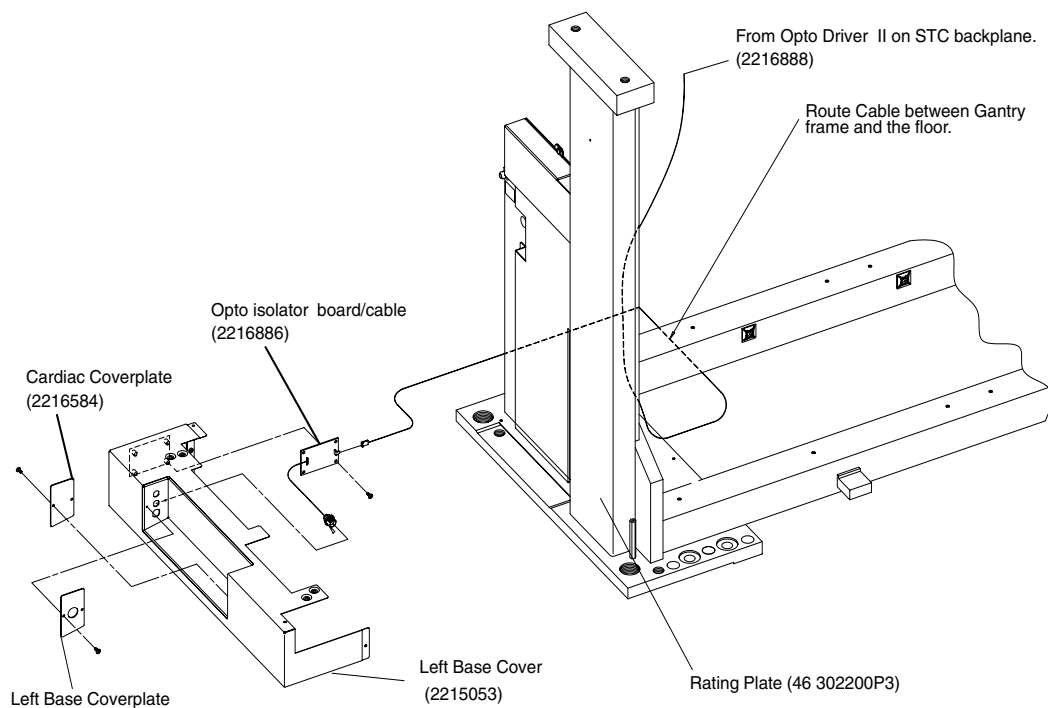


Figure 3-4 Route the Cardiac Cable In The Gantry Base

- 9.) Refer to [Figure 3-4](#).
- 10.) Route the cable connector under the gantry base, next to the floor, to avoid interfering with the covers.

Section 3.0

Install Opto-Isolator Board

Note: If the system has the SmartView Option, skip this section and proceed to Step 3.

- 1.) Remove the cardiac cover plate, 2216584, and left base plate cover, 2215053, from the upgrade kit.
- 2.) Attach the Cardiac cover plate to the left base plate cover.
- 3.) Refer to [Figure 3-4](#).
 - a.) Attach the cable from Opto-Isolator board, 2216886 to the middle hole on the gantry base cover. (If you attached the correct plate, you'll only see one hole.)
 - b.) Attach the Opto-Isolator board to the gantry base cover.
- 4.) Plug cable 2216888 into J1 of Opto-Isolator board, 2216886
Systems with the following Gantry Model Numbers have a factory-installed 2216886 cable. These gantries have the SmartScore Cardiac cable bundled and tied back to the existing harness.
 - 2183250-2
 - 2183255-2
 - 2119734-3
 - 2119732-4
- 5.) After you route the Cardiac cable into place in the gantry base, tie/wrap it to the existing cables.

Section 4.0

Optional Disk Drive

Note: Some SmartScore kits come with an optional disk drive for the OC to increase the customers image space. This optional disk drive can be installed at the same time as the gantry hardware is being installed. Also, the optional disk drive could be broken out from this installation and be installed at a later date and time depending on customer time constraints. If postponed, be aware the customers image space could be limited.

If this option is present, follow steps in the CT/i Options manual(2181737-100).

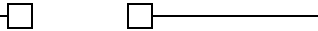
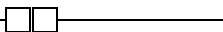
Chapter 4

System Functional Checks

Purpose: Use this procedure to ensure the proper installation and operation of the CACS EKG monitor. Install the CACS software package on the Advantage Windows work station before following this test procedure.

Section 1.0

Create Test EKG Waveform

- 1.) Turn on EKG monitor power:
 - a.) Turn ON the power switch on the front of the monitor.
 - b.) Turn ON the power switch on the back of the EKG monitor.
- 2.) Wait for the EKG monitor to pass its power-up self-tests.
- 3.) Locate the "seat-belt" GE Connect symbol in the bottom right corner of the monitor screen.
 - If necessary, disconnect the cable between the EKG monitor and gantry base.
 - Make sure the two halves of the symbol do not touch, which indicates a disconnected gantry cable. 
- 4.) Connect the cable from the left CT gantry base to the back of the EKG monitor. Make sure the "seat-belt" symbol closes. 
- 5.) Connect the gray EKG interface cable to the front of the monitor.
- 6.) Connect the black EKG lead cable to the interface cable by matching the colored ends of the flexible cable to the corresponding colored connectors on the EKG interface cable.
- 7.) Use the RA (white), RL (green), LL (red), and LA (black) connectors to attach the black flexible lead cable to the EKG heartbeat simulator .
- 8.) Turn on the EKG heartbeat simulator and set the heart rate to between 60 and 80 bpm. (beats per minute). Select the EKG output waveform and observe the waveform on the display.
- 9.) Locate the disk drive on the side of the EKG monitor, and insert a formatted, blank high density floppy disk into the drive opening.
- 10.) On the CT/i console, prescribe and acquire a 15 second helical scan of air.
- 11.) Record Patient ID number: _____
- 12.) Enter the same Patient ID into the EKG machine and press enter. The status box in the bottom right corner of the EKG display turns from black to green.
- 13.) Select CONFIRM and acquire the scan.
 - The status box on the EKG display turns from green to yellow when x-rays begin.
 - The status box on the EKG display turns from and from yellow to black when the scan completes.
- 14.) Wait for the EKG monitor to save the EKG data to the floppy disk.
- 15.) After the Patient ID screen appears, eject the disk from the drive.

Section 2.0

View the Test EKG Waveform

Note: Save the floppy you create in the following procedure. The floppy is used in the loading software in [Chapter 5](#).

- 1.) Take the disk to the Advantage Windows workstation and insert it into the disk drive.
- 2.) Start the CACS application.
- 3.) Select the INPUT tab
- 4.) Select UPDATE to locate the EKG file on the floppy disk.
- 5.) Select the EKG file name, and click OK.
- 6.) Select the ANALYSIS tab. Make sure the EKG waveform appears in green at the top of the screen.
- 7.) Return to the INPUT tab, select EEJECT, and remove the disk from the floppy drive.
- 8.) Quit the CACS application.

Chapter 5

SmartScore Option (AW Software) Installation

Section 1.0

Introduction: “New” vs. “Old” SmartScore

SmartScore Option reference manuals:

- **SmartScore Option Manual for CT/i & LightSpeed:**
http://www.ct.med.ge.com/tech_pubs/2152913/hsa-cti/cti/2224901.pdf
- **Applications Tips and Precautions:**
http://www.ct.med.ge.com/tech_pubs/2152913/qxi/2240919.pdf
- **Retrospective/Prospective Gating - Qxi:**
http://www.ct.med.ge.com/tech_pubs/2152913/qxi/2270001.pdf
- **Prospective Gating - LightSpeed Plus:**
http://www.ct.med.ge.com/tech_pubs/2152913/plus/2288566.pdf

SmartScore 3.5 software is installed and run on either an Advantage Workstation AW3.1_09 / AW4.0_05 or later Systems, as well as the AW4.1 current release.

SmartScore 3.3 software is installed and run only on Advantage Workstation AW3.1_09 and AW4.0_05.

The only SmartScore upgrade software intervention that MIGHT need to be done on the CT Scanner is the CT software options MOD procedure – see the CT installation or options manuals:

http://www.ct.med.ge.com/tech_pubs/www/techpubs/welcome.htm

Note: ALL the online links in this document are good places to start looking for documentation. However, due to organizational, server and website maintenance - online documents are occasionally moved, removed, and / or modified. The specific document, and / or link MIGHT change over time. This particular document will not be automatically updated to any online changes that occur. As is true with ALL online resources – it is up to the user to stay current with the available online documentation, its location, and its use.

Section 2.0

Summary of SmartScore Software Installation

Installing SmartScore software onto the AW Workstation can involve all or merely some of the following activities:

- Locating and having the SmartScore FDO and AW HOSTID based license key # and/or floppy if the SmartScore software is to be enabled at this time.
 - The SmartScore option should come with paperwork that includes an 8 to 10 digit license key number that was created based on the workstation's SmartScore FDO and the workstation's HOSTID or licenseld, if it is an AW4.1. Additionally, the SmartScore package should include portable media (i.e., floppy or CD-ROM) with the license key information on it. Both of these elements (hard copy license # and media) should exist, and at least one of them is necessary to load the SmartScore software.
 - If the SmartScore license key number is in paperwork form, it can be typed in during the SmartScore software installation in the 'Password Box' that appears during the

installation process.

- If there is a portable media license key that came with the SmartScore package, it can be used during the SmartScore installation to enable the SmartScore software. Instead of typing the license key in (leave a zero (0) in the password popup and click OK)
- Note: the media can be examined on a PC - or via the AW unix/linux mount commands - and the license key number can be visually read from the config.txt file, and written down for input during the SmartScore installation if necessary.
- Loading, and Enabling the CD-ROM SmartScore software package.
OR
- Loading (only) the CD-ROM SmartScore software package to the hard disk for future use. The load should commence if the license key box field is left as is, no license floppy is installed, and OK is clicked.

Section 3.0

SmartScore Software Installation Preparation

IMPORTANT BEST PRACTICE

- Note:
- AW2.0 up through AW4.0 uses a Configuration FLOPPY.
 - AW4.1 uses a configuration CD-ROM, instead of a floppy.

Create an AW 'CONFIG' floppy or CD-ROM NOW and AFTER the application software installation, for use as needed throughout this and other AW service activities.

Additionally, if there is a configuration floppy or CD-ROM on-site – with the package or freshly made – the configuration information can be read on a PC or in a command window on the AW, if need be:

- **PC**

Use any Windows-based text reader (e.g., Wordpad, Internet Explorer, MSWord) and open the config.txt file on the floppy/CD-ROM.

- Note:
- MS Notepad might NOT be able to read this file.

- **AW COMMAND WINDOW:**

```
su      -  
Password operator
```

- For a configuration FLOPPY disk (AW3.1 & 4.0), continue with:

```
mount -F pcfs /dev/diskette /floppy  
cd /floppy  
more config.txt  
cd ..  
umount /floppy
```

- For a configuration CD-ROM (AW4.1), continue with:

```
mount /mnt/cdrom  
cd /mnt/cdrom  
cd <backup directory>  
more config.txt  
cd /  
umount /mnt/cdrom
```


Section 4.0

SmartScore AW Software Installation Outline

AW3.1

- Advantage Workstation 3.1 SM for (Ultra 1, 2 or 10)
- Job Card entitled – Software Options Installation
- EASY-INSTALL details

AW4.0

- Advantage Workstation 4.0 SM for Ultra 60
- Job Card entitled – Software Options Installation
- EASY-INSTALL details

AW4.0P

- Advantage Workstation 4.0P SM for Ultra 80
- Job Card entitled – Software Options Installation
- EASY-INSTALL details

AW4.1

- Advantage Workstation 4.1 SM (Service Manual)
- Job Card entitled – Software Options Installation
- EASY-INSTALL details

The current revisions for all the manuals listed above can be viewed ONLINE – following the 'Documentation' links at:

<http://aw-ib.euro.med.ge.com/>

OR

<http://3.45.12.25/>

Section 5.0

Installation & Uninstallation Notes

- SmartScore no longer requires correctly installed AW printers/filmers to be identified and input during the SmartScore software installation. SmartScore will automatically detect successfully installed AW printers/filmers for selection within the application.
- **For AW4.0 and AW4.1:** After a successful SmartScore installation, the EasyInstall GUI window feedback might display:

Please REBOOT your workstation

This is because it MAY BE necessary to REBOOT THE WORKSTATION to avoid slight SmartScore GUI peculiarities with the 'window title bar' and the lower left 'progress bar.' If unsure about this, reboot the AW after the install to eliminate this possibility.

- IF, for whatever reason, the easy-install process will not work - there is a manual uninstall script that can be tried prior to re-installing:

```
cd /export/home/sdc/install
uninstall.sscore
```



NOTICE

The **uninstall.sscore** script **LOSES** the SmartScore configuration and logo files. **Save them first. See Test and Maintenance of SmartScore in Section 6.0.**

Save config files before running /export/home/sdc/install/uninstall.sscore

- **SmartScore 3.3** configuration files can only be saved within the SmartScore application, in the 'Preferences' screen, or by individually saving the unix files in the SmartScore file system - which is not detailed in this document.
- **SmartScore 3.5** and beyond configuration files can be saved the same as with SmartScore 3.3, but also with the standard AW SERVICES or ADMIN / Save Configuration utility.
- **Easy Install ERRORS:** If the following window pops up, you probably have selected the wrong media (Floppy /CD-ROM), or the concerned drive and or media is defective.

```
Notice
```

```
! Malformed package
```

```
Notice
```

```
! Error : Cannot access to media
```

Section 6.0

Test and Maintenance of SmartScore

6.1 SmartScore Application Configuration Save

In the SmartScore '**Preferences**' screen (within the SmartScore application), there will be a **SAVE/RESTORE** button. This is where the '*SmartScore only*' application configuration files will be saved and/or restored for use before and after a complete SmartScore uninstallation, via

```
~sdsc/install/uninstall.sscore.
```

Note: If a normal "easy install" SmartScore re-installation is performed without first using ~sdsc/install/uninstall.sscore, the SmartScore configuration is **AUTOMATICALLY** retained and restored during the new installation. This is true in SS3.3 and 3.5, and beyond:

- In SmartScore 3.3, if an AW LFC is scheduled or if ~sdsc/install/uninstall.sscore is going to be used, then the SmartScore 'Preferences' 'SAVE/RESTORE' process is the only method to **SPECIFICALLY** save the SmartScore 3.3 configuration information for future restoration.
- In SmartScore 3.5, the SmartScore specific preferences can similarly be saved (compared with 3.3). Additionally, however, in 3.5 these configuration files are also saved in the standard AW - SERVICES or ADMIN - Save Configuration utility. This would be for the purpose of returning the entire AW plus the SmartScore 3.5 application back to its configured state, after an AW service intervention - like reloading AW base software.

6.2 SmartScore Configuration SAVE/RESTORE Test

After successfully installing SmartScore:

- 1.) Customize the 'Preferences' screen for demographics, category, guide, hospital address, logo, and so on.
- 2.) Insert a PC formatted floppy disk (or appropriate portable media) into the AW computer.
- 3.) In the Preferences screen, SAVE the configuration files to the media.
- 4.) As a TEST: Prior to performing the restore procedure, open a unix shell and uninstall SmartScore using:
`~sdc/install/uninstall.sscore`



NOTICE

This intentionally loses all the SmartScore configuration and logo files.

- 5.) Re-install SmartScore using the same installation procedures as you did initially (Easy Install process).
- 6.) Insert the SmartScore configuration floppy
- 7.) Got to the Preferences screen
- 8.) Click on SAVE/RESTORE, and restore the configuration files from the media.

The operator preferences should be as they were when you had the operator customize them. If not, a reasonable troubleshooting process should be started to determine if the AW floppy mechanism is working OR if SmartScore is failing somehow - i.e.:

- Re-do entire procedure to make sure that a step was not incorrectly done.
- Notice whether or not the floppy was accessed during the save - did it sound like it?
- If so, read the floppy on a PC, or use the mount utilities on the AW platform to examine the floppy.
- If not, can it be accessed via other applications, (i.e fdfloppy)?
- Did the floppy get accessed when restoration was initiated?
- Are there any floppy/device type errors in the AW /var/adm/messages logfiles?
- Are there any SmartScore errors in ~sdc/logfiles/sscorelog?
- Re-load SmartScore, and try again?

Note: Instruct the operator to save the configuration to floppy on a regular basis – depending on how often the preferences items are changed. Of course, it is not necessary to `uninstall.sscore` – this was just to test the configuration save and restore function for the purposes of this installation.

SUMMARY

For SmartScore specific serviceability only (to save and restore the SmartScore specific preferences only), the SmartScore 'Preferences' 'SAVE/RESTORE' process is the only method in 3.3, and is the preferred method for both 3.3 & 3.5.

FOR AW SYSTEM SERVICEABILITY - IF AW BASE SOFTWARE IS TO BE RE-LOADED, THE SMARTSCORE 3.5 CONFIGURATION FILES WILL BE SAVED IF THE AW SERVICES or ADMIN SAVE CONFIGURATION PROCESS is used (as it should be before a LFC), AND CORRESPONDINGLY RESTORED AFTER THE RE-LOAD ALONG WITH THE `~sdc/Prefs` DATA.

Section 7.0

SmartScore 3.5 and Beyond, File Structure Information

The SmartScore file structure can be explored via the SmartScore directory (`/export/home/sdc/sscore`), and the various configuration and logo files can be discovered therein. Additionally, starting with SmartScore 3.5, the SmartScore configuration files will be saved in the `~sdc/Prefs` directory.

Section 8.0

SmartScore Maintenance Information (Declaration of Image Failure Error or File System Full Issues)

8.1 Information

SmartScore saves scored studies information in the `~sdc/sscore/studies` directory. The files stored here are called 'save state' files, because they retain the scoring details of the individual scoring session for that exam. In SmartScore 3.3 there is no automated process in place to limit the number of save state files that can be saved in this directory. So, even though these files are small, IF a large number of SmartScore studies are done on a regular basis, and/or IF other AW software processes are actively saving data – the size of the studies directory might fill up the `/export/home` partition. This can result in SmartScore failing with an error such as:

Declaration of Image Failure error

Or

The AW may not boot or it may crash

The preferred solution to this will be the implementation of DICOM STRUCTURED REPORTING, which will allow Image Data Sets to be updated or generated, and thereby be saved in the image database, and easily archived. However, neither SmartScore 3.3 or 3.5 have this implemented. Nevertheless, SmartScore 3.5 WILL have a fix built into it to systemically deal with save state file management. With respect to this, SmartScore 3.5 will start implementing a file system check during the EASY INSTALL process. During SmartScore installation, if there is insufficient file space for either the application files or the 'studies' files, a message will appear in the Easy Install GUI Window indicating the amount of file space needed, and potentially what to do about it.

EXAMPLE SMARTSCORE 3.5 INSTALLATION FILE SYSTEM CHECK MESSAGES:

- Software cannot be installed due to insufficient disk space in `/export/home...` SmartScore needs 70 MBytes (70000KBytes) of partition space to successfully install the application. Please remove “?software calculated value? MBytes” of data in the `/export/home` partition to allow SmartScore installation. Care should be taken when doing this, and the user may need to be consulted. If unsure about how to determine if files are needed, Please contact the OLC-AW team.
- Software cannot be installed due to insufficient disk space under the image partition `/export/home1...` SmartScore needs 60 MBytes (60000KBytes) of partition space to save its operational data files. Please remove “?software calculated value? MBytes” of data in the `/export/home1` partition to allow SmartScore installation. THE EASIEST SOLUTION MIGHT BE TO SIMPLY REMOVE “?software calculated value? IMAGES” FROM THE DATABASE. Care should be taken when doing this, and the user may need to be consulted. If unsure about how to determine file and image space characteristics, Please contact the OLC-AW team.

8.2 Procedure

8.2.1 For SmartScore 3.3 only (until it is replaced by 3.5)

Service intervention will be required to remove the excessive files from `~sdc/sscore/studies` preferably before a problem occurs. This will depend on how many SmartScore studies are done during a given period of time. A heavy usage site will need this intervention more frequently than a low usage site.

8.2.2 Service Procedure – from a command unix window

Check file system capacity:

```
{sdc@} [2] df -k
Filesystem      Kbytes      used      avail      capacity    Mounted on
/proc           0            0            0           0%          /proc
/dev/dsk/c0t0d0s0 819032      408347      353353      54%         /dev/fd
/dev/fd          0            0            0           0%         /dev/fd
/dev/dsk/c0t0d0s1 601592      291560      249873      54%         /export/home
swap            2004384      312         2004072      1%          /tmp
/dev/md/dsk/d0   58465360    20242528    37053525    36%         /export/home1
/dev/md/dsk/d1   1590073      9           1558263      1%          /export/cdr
{sdc@} [3]
```

When `/export/home` starts to reach a capacity of 100%, there will likely be a system problem. The goal is to keep this partition's capacity below 100%. Remember that other 'non-SmartScore' files in this partition can also cause and/or add to overcapacity. However, for the purposes of this SmartScore Maintenance Procedure, when the `~sdc/sscore/studies` directory reaches 800 files (roughly a file system capacity of 50M) all the files – or a subset of all the files – should be removed per the Removal of `~sdc/sscore/studies` save state files in 8.2.2.2.

8.2.2.1 Finding the capacity of the `~sdc/sscore/studies` directory

```
{sdc@} [1] cd ~sdc/sscore/studies
{sdc@} [2] du -k
141
{sdc@} [3]
```

<----- **Example Directory capacity (141K)**

There is really no magic number. An estimation of 800 files (50M) should be used in conjunction with the system's normal `df -k` output characteristics to allow a proactive service procedure that will hopefully avoid SmartScore caused file-system issues until SmartScore 3.5 replaces the installed base.

8.2.2.2 Removal of `~sdc/sscore/studies` save state files

- 1.) Find out from the customer what SmartScore studies in the studies directory file listing are now old, and not needed anymore. Probably NONE of them will be needed for future use. However, this is a site usage specific variable, and should be verified with the user before it is assumed.

A possible usage scenario would be where a follow-up study is planned on a patient, and therefore the save state information will be needed to compare to the last study. Theoretically, this should not be needed if a report was generated, but the customer could conceivably be using it this way.

- 2.) Find the savestate file listing in `~sdc/sscore/studies`
The files are named: `patientID_examID_seriesNumber`

Example:

```
{sdc@} [1] cd /export/home/sdc/sscore/studies
{sdc@} [2] ls -al
```

```
total 428
drwxr-xr-x  2 sdc    informix    512 Apr  8 10:44 ./
drwxr-xr-x 12 sdc    informix    512 Apr  8 10:35 ../
-rw-r--r--  1 sdc    informix  70720 Aug 21  2001 ##-##-##-##_####_#
-rw-r--r--  1 sdc    informix   71959 Aug 21  2001 #_#####_###
-rw-r--r--  1 sdc    informix   34158 Apr  8 10:42 NO NUMBER_####_#
-rw-r--r--  1 sdc    informix   36893 Apr  8 10:44 NO NUMBER_####_##
{sdc@} [3]
```

- 3.) Remove the identified unneeded files like the following example, or in blocks with the appropriate unix wildcard specifier:

Example:

```
{sdc@} [6]  rm  ##-##-##-##_####_#
rm: remove ##-##-##-##_####_# (yes/no)? y
{sdc@} [7]  rm  "NO NUMBER_####_#"
rm: remove NO NUMBER_####_# (yes/no)? y
{sdc@} [8]
```

Section 9.0

SmartScore Operation and Demo Exam Information

It is beyond the scope of this document to detail the correct operation of the SmartScore software application. See the operator's manual for identifying, and or creating the appropriate scan data sets to test SmartScore, and further detailed SmartScore functionality.

SmartScore should come with a CD-ROM of 'Demo' exams – or there will be SmartScore Demo exams on the AW Demo Exam CD-ROM set. These data sets can be 'archive restored' on to the AW, selected, and then launch SmartScore to see if it starts up and is available for use. Then reference the operator's manual for further detailed functionality.

Section 10.0

SmartScore Error Log Information

There is a log file on the system for SmartScore. Look here for installation and other suspected SmartScore failures:

```
/export/home/sdc/logfiles/sscorelog
```

Section 11.0

SmartScore Printing and Filming Information

The Post Script Printer and the DICOM FILMER (CAMERA) will be selected via a Menu in the SmartScore: REPORT – (PRINT OPT or OUTPUT)

Conversely, if there is no post script printer or no camera selectable in the REPORT - (PRINT OPT or OUTPUT) menus - it means that these have not been SUCCESSFULLY configured via the install.printers script and/or the AW - printer manager utility and/or - the device is simply OFF, DISCONNECTED, or BROKE? In other words:

Post Script printers and/or Dicom or Laser Cameras must be successfully configured and functional via their respective AW setup methods before SmartScore will recognize and use them.

Printers & Filmers, as well as their parameters can be removed, installed, or changed at any time. SmartScore will pick up the changes by simply quitting and restarting the SmartScore application. Re-installation of the SmartScore software is not necessary.

Additionally, SmartScore will PostScript Print images, as well as reports to a color postscript printer - if the printer is configured and selectable within SmartScore.

Note: SmartScore 3.2 on an AW4.0 would not print images if a color postscript printer was configured via install.printers. It had to be installed as a greyscale printer. This is not a factor anymore for SmartScore 3.3, and beyond.

SEE the AW service manual under the JOB Card 6 dealing with POSTSCRPT and DICOM PRINTER installation for detailed installation information.

Section 12.0

SmartScore “Special” Considerations

This section explains how to establish Research Mode Settings in SmartScore.

12.1 Read This First

This is not a “normal” installation or service procedure. IF, and ONLY IF the customer/radiologist requests - - the following is the procedure to configure specialized research scoring methods. In order to successfully do this, the customer must provide the following specific variables:

- 1.) The NAME of the method to appear on the report.
- 2.) The FILTER pixel connectivity setting.
- 3.) The SCORE CLASS
- 4.) The THRESHOLD

12.2 Procedure

- 1.) Open a unix command window and perform the following commands:

```
cd sscore          ENTER
cd bin             ENTER
ls                 ENTER
```
- 2.) At the top portion of the list you should see “config_sscore”. Enter the following command:

```
config_sscore      ENTER
```
- 3.) A window with TABS will open towards the top of the screen. The First Tab is “Display”. Below this will be THREE labeled fields:
 - THRESHOLD
 - DEFAULT LEVEL
 - DEFAULT WINDOW



NOTICE

The threshold value listed here must be the lowest value the site intends to use.

Example; If they intend on using the AJ 90 score as one of their available reports then they must enter 90 in this field, and all subsequent values must be larger.

- 4.) The Second Tab is "Score 1" - **DO NOT TOUCH.**
- 5.) The Third Tab is "Score 2" - **DO NOT TOUCH.**
- 6.) TABS Score 3-8 are available to configure.
Example Configuration:
 - a.) Click on tab "Score 3"
 - b.) **Name** - enter in the score name desired on the report - 8 characters MAX.
 - c.) **Filter** - this is pixel connectivity. The number given here is the number of additional pixels to be connected.
 - * A filter of 1= 2 pixel connectivity.
 - * A filter of 3=4 pixel connectivity.
 - * Typical value is 1 for GE scanners & 2 for Imatron.
 - * The maximum value you can enter in this field is 5.
 - d.) **Score Class** - choices are:
 - * SteppedScorer - all AJ reports fall into this class
 - * LinearScorer - invokes Linear scoring algorithm
 - * VolumeScorer - measures volume of the calcium in mm³
 - * MassScorer - generates a score value proportional to the true mass of calcium.
 - e.) **Threshold** - User defines - see NOTICE in step 3.
 - f.) Continue this process for Score 4 - 8, if desired.
- 7.) Click on SAVE
- 8.) Click on QUIT
- 9.) Type **exit** and return to patient browser
- 10.) RESTART BROWSER to save new changes.

12.3 Check Scoring Configuration

- 1.) Launch SmartScore and go into the PREFERENCES Screen.
 - 2.) Select SCORE TABLE from the list of items in the white vertical window.
 - 3.) In the ALGORITHM WINDOW, you should see all the scoring methods that are currently configured - including the one(s) that you setup with the `config_sscore` process.
The scoring methods should have a 'CHECK BOX' that - when checked - designates the scoring method to appear in the REPORT output.
 - Make sure that the scoring method(s) that you configured with the `config_sscore` process appear in the ALGORITHM WINDOW.
If it (or they) do not, it could be that perhaps the Browser was not restarted after the `config_sscore` activity.
 - Presumably, the user would like the newly configured scoring method(s) to appear in the REPORT output. Make sure that there is a CHECK MARK in the box corresponding to the new scoring method(s). DO NOT CHANGE THE CHECK BOX STATUS OF ANY OF THE SCORING METHODS THAT YOU DID NOT CONFIGURE - unless the user specifically asks you to.
For further verification, a scoring report can be generated to verify that the scoring method(s) appear in the report output.
- Congratulations, you have added the desired research modes.

Chapter 6

SmartScore (System) Interconnect

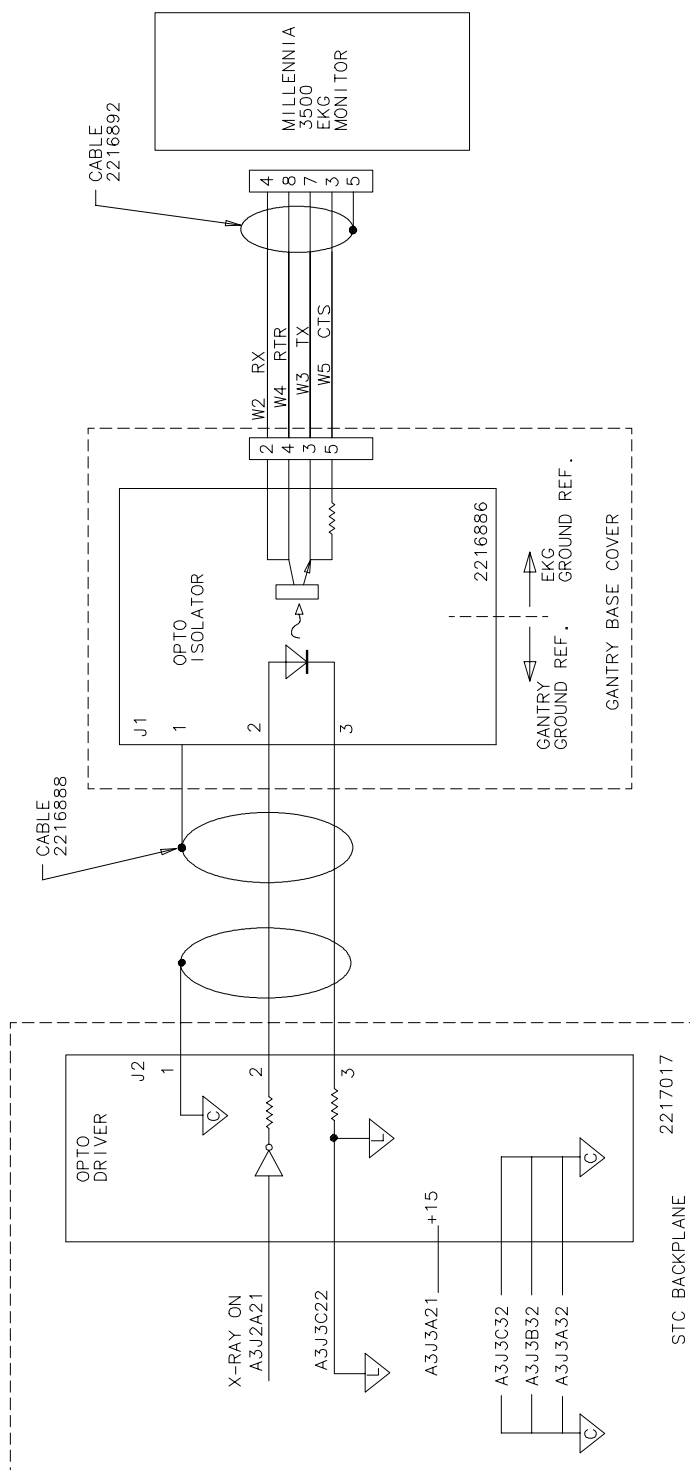


Figure 6-1 SmartScore Option Interconnect

Chapter 7

Troubleshooting

Testing without EKG: *Please refer to System Diagram, in Figure 6-1 on page 41.*

The interface integrity can be evaluated using an ohm-meter when the EKG is not attached to the cable "D" connector at the monitor. X-Ray ON will be Low when not scanning and will be High when scanning.

- 1.) Make sure cable 2216892 is connected to the gantry base
- 2.) Measure the voltages at the connector end attached to the monitor.

The following ohm-meter readings would be expected at the "D" connector in a normal installation:

Condition (SIGNAL)	Connection			Measured Value
	(Positive)	To	(Negative)	
X-RAY ON (Low)	Pin 5	to	Gantry Ground	open
X-RAY ON (Low)	Pin 3	to	Gantry Ground	open
X-RAY ON (Low)	Pin 4	to	Pin 8	short
X-RAY ON (Low)	Pin 3	to	Pin 7	2 k ohms
X-RAY ON (Low)	Pin 8	to	Pin 7	< 3 k ohms
X-RAY ON (High)	Pin 5	to	Gantry Ground	open
X-RAY ON (High)	Pin 3	to	Gantry Ground	open
X-RAY ON (High)	Pin 4	to	Pin 8	short
X-RAY ON (High)	Pin 3	to	Pin 7	2 k ohms
X-RAY ON (High)	Pin 8	to	Pin 7	> 20 k ohm

Table 7-1 Nominal Values - On cable 2216892 at Monitor Connection Side

**The Invivo 3500CT EKG machine is serviced by Invivo.
For Service or Questions in the U.S., please call 1-800-331-3220**



GE MEDICAL SYSTEMS

***GE MEDICAL SYSTEMS-AMERICAS: FAX 414.544.3384
P.O. BOX 414; MILWAUKEE, WISCONSIN 53201-0414, U.S.A.***

***GE MEDICAL SYSTEMS-EUROPE: FAX 33.1.40.93.33.33
PARIS, FRANCE***

GE MEDICAL SYSTEMS-ASIA: FAX 65.291.7006